

Mathematical Aspects of (Gauge) Field Theory

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Abstract

This is a note on the mathematical basis of gauge theory. I intend to follow the renowned textbook [1], which will provide an introduction to the mathematical tools of: homology group, cohomology group, fibre bundle, connection, chern class and index theorem which are widely used in gauge theory.

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1 Homology Group

Bibliography

- [1] M. Nakahara, *Geometry, Topology and Physics, Second Edition*. Taylor & Francis, 2003. doi: [10.1201/9781420056945](https://doi.org/10.1201/9781420056945).